

Participant Instructions for Alteron User Study

Thank you for participating in this study.

This study evaluates the usability and interpretability of Alteron, a behavioral regression testing tool for NLP classifiers. You will run a prepared study bundle, inspect the generated artifacts, and complete a Google Form.

Estimated Time

20-30 minutes

What You Need

- A terminal
- Access to the provided Alteron study bundle
- The Google Form link
- Basic command-line familiarity

You do **not** need to train a model, modify code, or understand the full repository.

Study Goal

You will use Alteron in a prepared scenario to -

1. run the tool on a candidate model,
2. inspect the generated outputs,
3. determine whether release would be blocked,
4. identify the cause of the blocking result, and
5. explain what the outputs mean.

About Alteron

Alteron is a behavioral regression testing tool for NLP classifiers. Its goal is to check whether a new model version preserves expected behavior when compared with an accepted baseline model. In addition to ordinary task accuracy, Alteron evaluates how model behavior changes under controlled input transformations called metamorphic relations (MRs). These checks are designed to help teams decide whether a candidate model should be released in a CI/CD workflow.

Key Terms

Behavioral regression. A behavioral regression occurs when a candidate model behaves worse than the baseline under one or more metamorphic tests, even if ordinary task accuracy does not decrease.

Behavior regresses even though accuracy improves/remains unchanged. A model can improve standard accuracy on original inputs while still becoming less robust or less behaviorally consistent under transformed inputs. Alteron is designed to detect this kind of change.

Frozen corpus. A frozen corpus is a fixed set of validated source and follow-up test pairs reused across model versions so that changes in results come from model differences rather than test-set differences.

Snapshot. A snapshot is the recorded behavior of a model on the frozen corpus, including predictions, confidence scores, and MR pass/fail outcomes.

Regression report. A regression report summarizes the comparison between the baseline and candidate for each MR. It includes the MR identifier, the change in source accuracy, the change in MR pass rate on the matched subset, whether a behavioral regression was flagged, and whether the result blocks release.

Setup

You will receive a prepared bundle inside the repository under `user-study/`

The bundle already contains -

- a tiny frozen corpus,
- a prebuilt baseline snapshot,
- a mock candidate model,
- a one-command wrapper script.

Important Note

Please use the provided study materials exactly as they are. Do not edit files or change the command.

Step-by-Step Instructions

Step 1 - Open the repository

Open a terminal in the root of the provided Alteron repository.

Step 2 - Run the study

Run these command (linux or mac):

```
./user-study/setup_study.sh
```

```
./user-study/run_study.sh
```

This command executes the prepared Alteron study scenario.

For windows, check out the PARTICIPANTS_INSTRUCTIONS.md file inside the folder.

Expected Outcome

The command is expected to complete and produce output files under:

```
user-study/study_output/
```

If the command prints a failing CI result, that is part of the study scenario and does not mean you made a mistake.

Step 3 - Inspect the outputs

After the run completes, inspect these files:

```
user-study/study_output/regression_reports/regression_report_stable_to_candidate.csv
```

```
user-study/study_output/ci_summary.json
```

You may also inspect other files in:

```
user-study/study_output/ if helpful.
```

Step 4 - Complete the Google Form

Answer the Google Form questions based on:

- What you observed while running the study,
- The contents of the generated output files,
- Your interpretation of the tool's result.

What you should not do

- Do not train models
- Do not regenerate corpora
- Do not modify the repository
- Do not debug the internals unless the instructions explicitly tell you to
- Do not spend time trying to optimize or fix the prepared scenario

If something goes wrong

If the script does not run:

1. note the issues briefly,
2. continue to the Google Form,
3. report the issues in the relevant question.

What we are evaluating

1. Whether the workflow is understandable
2. Whether the outputs are easy to interpret
3. Whether Alteron seems useful in a CI/CD or model-maintenance setting.

Reminder

The most important output files for this study are:

user-study/study_output/regression_reports/regression_report_stable_to_candidate.csv

user-study/study_output/ci_summary.json

Thank You

Once you finish, please submit the Google Form.